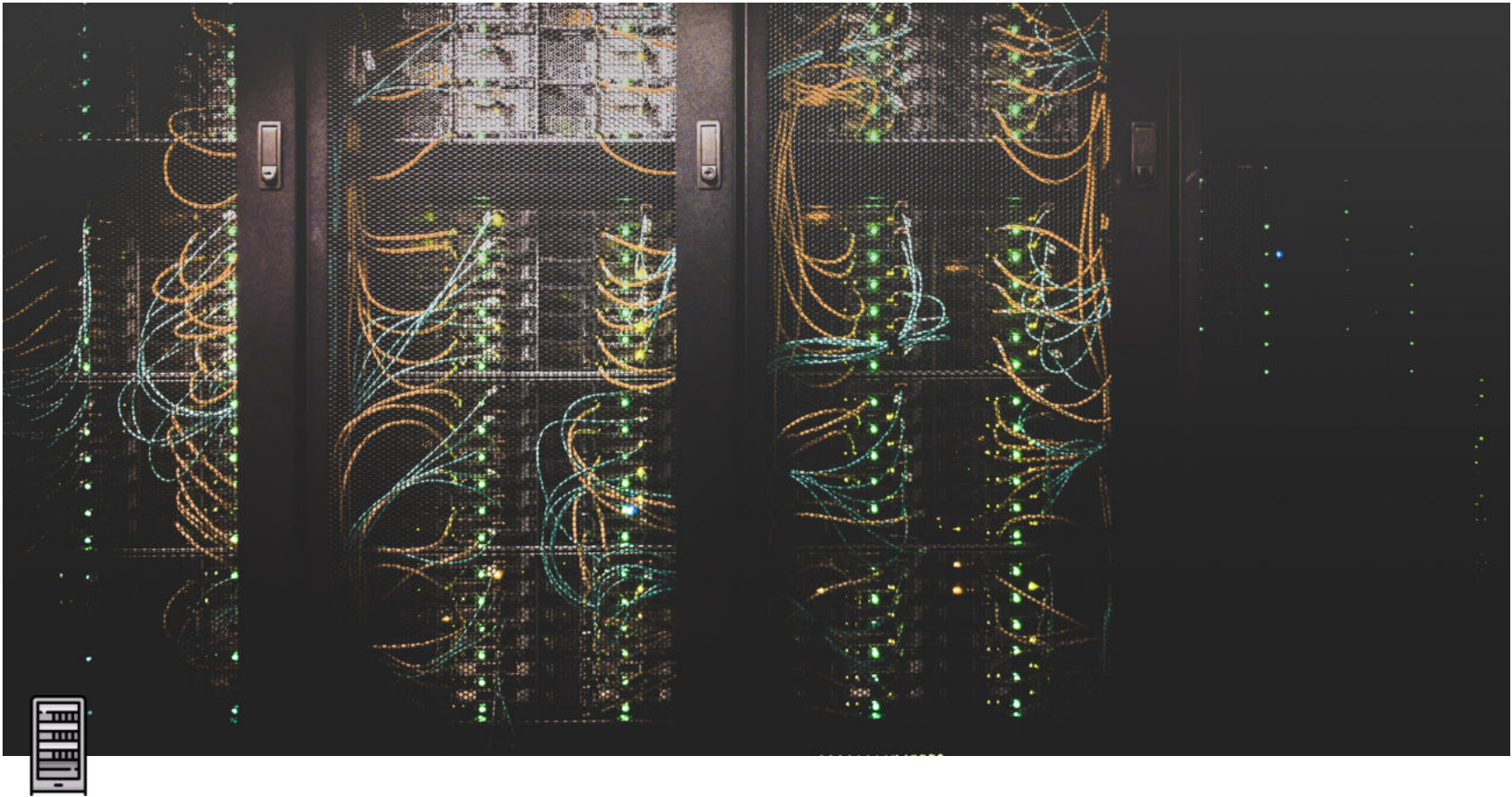




Shell variables

Type	Lecture
Date	@December 22, 2021
Lecture #	2
Lecture URL	https://youtu.be/pPRge8Yxbso
Notion URL	https://21f1003586.notion.site/hell-variables-7c2117f90cee4aa7aec3ee410038f32a
Week #	2

echo

echo command prints a string or a environment variable

Example:

```
echo $HOME
echo Hello, World
```

Frequently used shell variables

- \$USERNAME
- \$HOME
- \$HOSTNAME
- \$PWD
- \$PATH

Special shell variables

- \$0 → Name of the shell
- \$\$ → Process ID of the shell
- \$? → Return code of previously run program
- \$- → Flags set in the bash shell

Process control

- Use of & to run a job in the background
- fg

- `coproc`
- `jobs`
- `top`
- `kill`

Program exit codes

- 0 → success
- 1 → failure
- 2 → misuse
- 126 → command cannot be executed
- 127 → command not found
- 130 → processes killed using `Ctrl + C`
- 137 → processes killed using `kill -9 <pid>`

Flags set in `bash`

- `h` → locate and hash commands
- `B` → brace expansion enabled
- `i` → interactive mode
- `m` → job control enabled
- `H` → ! style history substitution enabled
- `s` → commands are read from stdin
- `c` → commands are read from arguments

`echo`

```
echo Hello World
echo "How do you do?"
```

it's a start

We can enclose the string in quotes as well

Make sure to match the quotes properly

Some common `echo` commands

```
echo $USERNAME
echo $USER
echo $PWD
echo $HOME
echo $HOSTNAME
echo $PATH
```

We can enclose the shell variable (marked as \$) in double quotes, and it'll replace the value

However, if we use single quote around the shell variable, the shell variable is printed as it is

We can also escape the shell variable by using a backslash in front of it

Example → `echo "User is \ $USERNAME"` will display `User is $USERNAME`

To view all the shell variables

```
printenv
env
```

`printenv` prints all or part of the environment

```
printenv HOME
```

set

set allows you to change the values of shell options and set the positional parameters, or to display the names and values of shell variables. ([Source](#))

To escape any aliases set already

```
\<alias>

# Example
\ls
\date
```